

Part C/MSc

Bayes Methods

Mock 2 2018

Do not turn this page until you are told that you may do so

1. A field is divided into an $m \times n$ grid of square cells. The number $y_{i,j}$ of buttercups in each cell (i, j) is counted. The observation model is

$$y_{i,j} \sim \text{Poisson}(\mu_{i,j}), \quad i = 1, \dots, m, \quad j = 1, \dots, n.$$

Suppose we parameterise $\mu = \phi\psi^T$ as $m \times n$ outer product, where $\phi = (\phi_1, \dots, \phi_m)^T$, $\phi \in [0, \infty)^m$ and $\psi = (\psi_1, \dots, \psi_n)^T$, $\psi \in [0, \infty)^n$ are column vectors of positive parameters.

Thinking of the cells in the matrix μ as a lattice, denote by $N_{i,j}$ the set of labels (a, b) , $a \in 1 : m$, $b \in 1 : n$ for cells adjoining cell (i, j) (interior cells have 4 neighbors, and boundary cells 3, except at the corners where each cell has 2 neighbors). Consider the joint prior density

$$\pi(\phi, \psi) \propto \exp(-\alpha\bar{\phi} - \beta\bar{\psi}) \exp(-\theta \sum_{i=1}^m \sum_{j=1}^n \sum_{(a,b) \in N_{i,j}} |\phi_a \psi_b - \phi_i \psi_j|),$$

where $\bar{\phi} = m^{-1} \sum_{i=1}^m \phi_i$, $\bar{\psi} = n^{-1} \sum_{j=1}^n \psi_j$ and α, β and θ are fixed positive prior hyperparameters.

(a)

- (i) Show that $\pi(\phi, \psi)$ is a proper prior.
- (ii) Write down the posterior distribution $\pi(\phi, \psi | y)$ for ϕ and ψ in terms of the model elements.
- (iii) Write down a simple Metropolis Hastings algorithm targeting $\pi(\phi, \psi | y)$.
- (iv) Let $\chi = (\phi, \psi)$ and $\chi' = (\phi', \psi')$. Suppose the transition density for your algorithm is $K(\chi, \chi')$. Show that K satisfies detailed balance with respect to π . Why is this important?

(b)

- (i) Let $u \sim g(u)$ be a random variable with density g and suppose $(\chi, u) \in \Omega$. Let $f : \Omega \rightarrow \Omega$ be an invertible differential function and suppose that if $(\chi', u') = f(\chi, u)$ then $(\chi, u) = f(\chi', u')$. Given χ we generate an MCMC proposal by simulating $u \sim g(u)$, computing $(\chi', u') = f(\chi, u)$ and taking χ' as our proposal. Write down the acceptance probability for this proposal.
- (ii) Suppose $u_1 \sim U(0.5, 2)$, $u_2 \sim U(0.5, 2)$ and $(\chi', u') = f(\chi, u)$ with $u = (u_1, u_2)$ and

$$\begin{aligned} \phi' &= \bar{\phi} + u_1(\phi - \bar{\phi}) \\ \psi' &= \bar{\psi} + u_2(\psi - \bar{\psi}). \end{aligned}$$

Show that $(\chi, u) = f(\chi', u')$ with $u' = (u'_1, u'_2) = (1/u_1, 1/u_2)$.

- (iii) Show that the acceptance probability is

$$\alpha(\phi', \psi' | \phi, \psi) = \min \left\{ \frac{\exp(-\theta \sum_{i,j} \sum_{(a,b) \in N_{i,j}} |\phi'_a \psi'_b - \phi'_i \psi'_j|)}{\exp(-\theta \sum_{i,j} \sum_{(a,b) \in N_{i,j}} |\phi_a \psi_b - \phi_i \psi_j|)} u_1^{m-3} u_2^{n-3} \right\}$$

noting any further conditions on ϕ', ψ' .

Hint: Suppose A is an invertible square matrix and u, v are column vectors.

$$\det(A + uv^T) = (1 + v^T A^{-1} u) \det(A).$$

This acceptance probability should read $\min\{1, \dots\}$. Also, the likelihood of the data is missing - a factor $p(y|\phi', \psi')/p(y|\phi, \psi)$ missing before u_1 .

2. (a) Let $S = (S_1, \dots, S_K)$ be a partition of $1 : n = \{1, 2, \dots, n\}$.
- (i) Specify the Chinese Restaurant process for arrivals $1, 2, \dots, n$.
 - (ii) Let $P_{CRP}(S)$ equal the probability the outcome of the CRP is the random partition S . Show that

$$P_{CRP}(S) = \frac{\Gamma(\alpha)\alpha^K}{\Gamma(\alpha+n)} \prod_{k=1}^K \Gamma(n_k)$$

[Note that $\Gamma(\alpha+n) = \Gamma(\alpha) \prod_{i=1}^n (\alpha+i-1)$.]

- (iii) Show that $P_{CRP}(S)$ does not depend on the order of the arrivals.
 - (iv) Let $i_1, i_2, i_3 \in 1 : n$ be three fixed labels. What is the probability that i_1, i_2, i_3 are in the same partition set?
- (b) Consider a mixture of normal densities with a fixed number M of components, Dirichlet distributed mixture component weights $w = (w_1, \dots, w_M)$, and a prior $\pi(\theta^*)$ for the mixture component parameters $\theta^* = (\theta_1^*, \dots, \theta_M^*)$:

$$\begin{aligned} w &\sim \text{Dirichlet}(\alpha_1, \dots, \alpha_M) \quad \text{with } \alpha_M = \alpha/M \text{ for } \alpha > 0 \text{ fixed;} \\ z_i &\sim \text{Multinom}(w), \quad i = 1, \dots, n; \\ \theta_m^* &\sim \pi(\theta_m^*), \quad m = 1, \dots, M. \end{aligned}$$

Here $z_i \sim \text{Multinom}(w)$, $i = 1, \dots, n$ means that $z_i = m, m \in \{1, \dots, M\}$ with probability w_m . In this model $z_i \in \{1, \dots, M\}$ is the label of the cluster to which y_i belongs. The observation model is

$$y_i \sim f(y_i; \theta_{z_i}^*), \quad i = 1, \dots, n.$$

Suppose the list $z = (z_1, \dots, z_n)$ of cluster labels contains $K \leq M$ unique values $m_1, \dots, m_K$. For $k = 1, \dots, K$ let $S_k = \{i : z_i = m_k, i = 1, \dots, n\}$. Let $S = (S_1, \dots, S_K)$. We write $S = S(z)$ for the partition determined from z in this way.

- (i) Write down the posterior for $\theta^*, z, w|y$ in terms of the model elements.
- (ii) Calculate the marginal prior probability $\pi_z(z) = \int \pi_{z|w}(z|w)\pi_W(w)dw$ for a set of cluster labels.
- (iii) For $k = 1, \dots, K$, let $n_k = |S_k|$ give the number of items in cluster k . Let

$$P(S) = \sum_{z:S(z)=S} \pi_z(z)$$

denote the prior distribution over partitions. Show that

$$P(S) = \frac{\Gamma(\alpha)}{\Gamma(\alpha/M)^K} \frac{M!}{(M-K)!} \frac{\prod_{k=1}^K \Gamma(\alpha/M + n_k)}{\Gamma(\alpha+n)}.$$

- (iv) Show that the prior distribution over partitions converges to a CRP as $M \rightarrow \infty$.
- (v) Write down the distribution to which the marginal posterior distribution of $\theta^*, S|y$ converges in the limit as $M \rightarrow \infty$.

3. Let A be the points in the unit square $[0, 1]^2$. For $i = 1, \dots, n$, let $x_i = (x_{i,1}, x_{i,2})$ be points $x_i \in A$ and let $x = \{x_1, \dots, x_n\}$ be a set of n points. The number and locations of the points are distributed according to a Strauss point process with parameters $\lambda > 0, 0 \leq \gamma \leq 1$ and $R \geq 0$. In particular, let $c(x; R)$ be the function

$$c(x; R) = \sum_{i=1}^{n-1} \sum_{j=i+1}^n \mathbb{I}_{|x_i - x_j| < R}$$

counting the number of R -close pairs of points in x . Let Ω_n be the set of distinguishable point patterns $x = \{x_1, \dots, x_n\}$ in A and let $\Omega = \cup_{n=0}^{\infty} \Omega_n$. The distribution of $x|\lambda, \gamma, R$ is

$$p(x|\lambda, \gamma, R) = \xi(\lambda, \gamma, R) \gamma^{c(x, R)} \lambda^n,$$

where $\xi(\lambda, \gamma, R)$ is a normalising constant ensuring the density is normalised to one over Ω .

(a) [5 marks]

- Give an expression for the normalising constant $\xi(\lambda, \gamma, R)$ for $p(x|\lambda, \gamma, R)$.
- Suppose $\gamma = 1$. Show, by integrating $p(x|\lambda, \gamma, R)$ over $x \in \Omega_n$ at fixed n , that the number of points n in the random set x has a Poisson distribution.

(b) [8 marks] Suppose x is observed and we wish to estimate λ, γ, R .

- Write down the posterior density $\pi(\lambda, \gamma, R|y)$ giving the density in as much detail as you can but without specifying priors.
- Suppose an ABC algorithm approximately targeting $\pi(\lambda, \gamma, R|y)$ and based on summary statistics $S(x)$ is available. The ABC algorithm yields samples $(\lambda^{(t)}, \gamma^{(t)}, R^{(t)})$, $t = 1, 2, \dots, T$ with associated statistics $S^{(t)}$. Explain how to make a regression adjustment to the sampled $\gamma^{(t)}$ -values. What is the purpose of this adjustment and how is it justified?
- What exact distribution does your ABC algorithm target? Briefly interpret the approximation being made in ABC.

(c) Consider a reversible jump algorithm simulating the Strauss process $x \sim p(x|\lambda, \gamma, R)$. The algorithm is structured as follows. Suppose $X_t = x$ with $x = (x_1, \dots, x_n)$ so point pattern x has n points in it.

To add a point pick a new point $z \sim A$ (throw a dart at A). Let $x' = x \cup z$ and $n' = n + 1$. The number of R -close pairs goes up to $c(x')$ and the number of points goes up to $n + 1$.

To delete a point, choose a point $x_i \in x$ at random from $i = 1, 2, \dots, n$. Let $x' = x \setminus x_i$ and $n' = n - 1$. The number of R -close pairs goes down to $c(x')$ and the number of points goes down to $n - 1$.

- Calculate the acceptance probability $\alpha(x'|x)$ when x' differs from x by the addition of one point.
- Calculate the acceptance probability $\alpha(x'|x)$ when x' differs from x by the deletion of one point.
- Outline a reversible jump MCMC algorithm targeting $p(x|\lambda, \gamma, R)$.

$$\delta/S \sim \mu_S - 1 \in S$$

BMIQ Mode 2

Q1 (a) (i). ^{show} $\int \pi(\phi, \psi) d\phi^m d\psi^n < \infty$

but $\exp(-\theta \dots) \leq 1$ and

$$\int e^{-\frac{\alpha}{m} \sum_{i=1}^m \phi_i - \beta \sum_{j=1}^n \psi_j} d\phi d\psi = \left(\frac{m}{\alpha}\right)^m \left(\frac{n}{\beta}\right)^n < \infty$$

so integral is finite so prior is proper.

(ii). $\pi(\phi, \psi | y) \propto L(\phi, \psi; y) \pi(\phi, \psi)$

$$L(\phi, \psi; y) \propto \prod_{i=1}^m \prod_{j=1}^n (\phi_i \psi_j)^{y_{ij}} e^{-\phi_i \psi_j}$$

(iii) $\phi^{(t)} = \phi, \psi^{(t)} = \psi$ (simplest ok here)

(1) $\phi'_i \sim U(\phi_i - \delta, \phi_i + \delta) \quad i=1 \dots m$

$\psi'_j \sim U(\psi_j - \delta, \psi_j + \delta) \quad j=1 \dots n$

$I_+ = \prod_{i=1}^m \prod_{j=1}^n \mathbb{I}_{\phi_i > 0} \mathbb{I}_{\psi_j > 0}$

(2) w.p. $\alpha(\phi', \psi' | \phi, \psi) = \min \left\{ 1, \frac{\pi(\phi', \psi' | y)}{\pi(\phi, \psi | y)} \right\} \times I_+$

Set $\phi^{(t+1)} = \phi', \psi^{(t+1)} = \psi$ else $\phi^{(t+1)} = \phi, \psi^{(t+1)} = \psi$.

[Remark, as this may be inefficient but we don't have enough info to know - depends on λ also - also +ve testing could be done before evaluating α , but above is min acceptable].

(iv) Above has $q(x' | x) = q(x | x')$

$$k(x, x') = q(x' | x) \alpha(x' | x)$$

Detailed balance is

$$\pi(x'|y) K(x, x') = \pi(x|y) K(x', x)$$

$$\Rightarrow \pi(x|y) \alpha(x'|x) = \pi(x'|y) \alpha(x|x')$$

WLOG assume $\pi(x'|y) \geq \pi(x|y) \Rightarrow \alpha(x'|x) = 1$

$$\Rightarrow \pi(x|y) = \pi(x'|y) \times \min\left(1, \frac{\pi(x|y)}{\pi(x'|y)}\right) \\ = \pi(x|y) \quad \text{since } \frac{\pi(x|y)}{\pi(x'|y)} \leq 1$$

So DB holds. Important for since sufficient for $\pi(x|y)$ stationary dbn. with irreducibility leads to ergodicity.

1 (b) i) "Write down"

$$\alpha(x'|x) = \min\left\{1, \frac{\pi(x'|y)}{\pi(x|y)} \frac{g(u')}{g(u)} \left| \frac{\partial(x', u')}{\partial(x, u)} \right| \right\}$$

$$\left[\text{ensures } \pi(x|y) g(u) \alpha(x', u' | x, u) = \pi(x'|y) g(u'(u, x)) \left| \frac{\partial(x', u')}{\partial(x, u)} \right| \alpha(x, u | x', u') \right]$$

(ii) must show if $u'_1 = 1/u_1$, $u'_2 = 1/u_2$ then

$$\phi = \bar{\phi}' + u'_1(\phi' - \bar{\phi}')$$

$$\psi = \bar{\psi}' + u'_2(\psi' - \bar{\psi}')$$

But $\bar{\psi} = \bar{\phi}$ ($\sum_i (\psi_i - \bar{\psi}) = 0$ etc) so rearranging

$$\phi = \bar{\phi} + \frac{1}{u_1}(\phi' - \bar{\phi})$$

$$= \bar{\phi}' + u'_1(\phi' - \bar{\phi}') \quad \text{similarly } \psi.$$

The likelihood ratio is missing in Eqn (*) - a factor $L(\phi', \psi'; y) / L(\phi, \psi; y) = p(y|\phi', \psi') / p(y|\phi, \psi)$ - present in the first line but dropped in the second. Can add $\ell(\phi', \psi'; y) - \ell(\phi, \psi; y)$ to H , with ℓ the log-likelihood.

The expression for $L(\phi, \psi; y)$ is given in 1a(ii) and does not need to be repeated.

2 (b) III Since $\bar{\phi} = \bar{\phi}'$ $\bar{\psi} = \bar{\psi}'$ $e^{-\alpha \bar{\phi} - \beta \bar{\psi}}$ cancels,

$$\alpha(\phi', \psi' | \phi, \psi) = \min \left\{ 1, \frac{\pi(\phi', \psi') L(\phi', \psi'; y) g(u')}{\pi(\phi, \psi) L(\phi, \psi; y) g(u)} \left| \frac{\delta(\phi', \psi', u')}{\delta(\phi, \psi, u)} \right| \right\}$$

$$= \min \left\{ 1, \frac{e^{-\theta H(\phi', \psi')}}{e^{-\theta H(\phi, \psi)}} \left| \frac{\delta(\phi', \psi', u')}{\delta(\phi, \psi, u)} \right| \right\}$$

$$H(\phi, \psi) = \sum_{i,j} \sum_{(a,b) \in N_{ij}} |\phi_i \psi_j - \phi_a \psi_b|$$

$$J =$$

	ϕ'	ψ'	u_1'	u_2'
ϕ	A	0	0	0
ψ	0	B	0	0
u_1	$\phi - \bar{\phi}$	0	$-1/u_1$	0
u_2	0	$\psi - \bar{\psi}$	0	$-1/u_2$

$$\phi_i' = \bar{\phi} + u_1(\phi_i - \bar{\phi})$$

$$\frac{\partial \phi_i'}{\partial \phi_i}$$

$$A = \begin{bmatrix} \diagdown & & \\ & d_i & \\ 0_{ij} & & \diagup \end{bmatrix}$$

$$\text{diag } d_i = \frac{1}{m} + u_1(1 - \frac{1}{m})$$

$$\text{off d. } 0_{ij} = \frac{1}{m} - \frac{u_1}{m} = \Delta$$

$$\frac{\partial \phi_i'}{\partial \phi_i}$$

$$= \begin{bmatrix} \diagdown & & \\ & u_1 + \Delta & \\ \Delta & & \diagup \end{bmatrix}$$

$$\Delta = \frac{1}{m} - \frac{u_1}{m}$$

$$A = u_1 I_m + \overset{\textcircled{u}}{\Delta} \overset{\textcircled{v^T}}{11^T} \quad \text{from hint}$$

$$\begin{pmatrix} u_1 & & 0 \\ & \ddots & \\ 0 & & u_1 \end{pmatrix} + \begin{pmatrix} 1 & \dots & 1 \\ \vdots & \ddots & \vdots \\ 1 & \dots & 1 \end{pmatrix} \times \Delta$$

$$\det(A) = \left(1 + \overset{\nearrow v^T}{\Delta 1^T} \overset{\nwarrow \textcircled{1}^{-1}}{\frac{I}{u_1}} 1 \right) u_1^m \leftarrow \det \textcircled{1}$$

$$= \left(1 + \frac{m}{u_1} \Delta \right) u_1^m$$

$$= \left(1 + \frac{m}{u_1} \left(\frac{1}{m} - \frac{u_1}{m} \right) \right) u_1^m$$

$$= \left(1 + \frac{1}{u_1} - 1 \right) u_1^m$$

$$= u_1^{m-1} \quad (\det(B) = u_2^{n-1})$$

$$|\det(J)| = |\det(A)| |\det(B)| \cdot \frac{1}{u_1^2} \frac{1}{u_2^2}$$

$$= u_1^{m-3} u_2^{n-3}$$

Sub this in * get ans.

Mock 2

Q2 1st customer sits at table 1. $S_1 = \{1\}$.

(i) After i^{th} arrival tables $S_1 \dots S_{k(i)}$

occupied $n_j^{(i)} = \# \text{ at table } j \in \{1, \dots, k(i)\}$

if st assigned table j w.p. $\frac{n_j^{(i)}}{\alpha + i}$

ii " new table w.p. $\frac{\alpha}{\alpha + i}$

Outcome is partition $S_1 \dots S_k$ of $1 \dots n$.

2(a) is led notes

2(a)(iii) $P_{\text{CSP}}(S)$ does not depend on

labels only on totals n_k so

if we see permuted order in which customers

arrive permuted outcome occurs with

same prob.

(iv) $\Pr(i_1, i_2, i_3 \text{ in same set})$

$$= \Pr(1, 2, 3 \text{ " " "})$$

$$= \frac{1}{\alpha+1} \times \frac{2}{\alpha+2}$$

(b) (i)

$$\pi(\theta^*, z, w|y) \propto \pi(w) \pi(z|w) \prod_{m=1}^M \pi(\theta_m^*) \\ \times \prod_{i=1}^n f(y_i; \theta_i^*)$$

$$\pi(w) = \frac{\prod_i P(x_i)}{\prod_i P(\alpha_i)} \prod_{i=1}^M w_i^{\alpha_i - 1}$$

$$\pi(z|w) = \prod_{m=1}^M w_m^{\hat{n}_m} \quad (\text{as for } w_i)$$

$$\hat{n}_m = \sum_{i=1}^n \mathbb{I}_{z_i = m} \quad \# \text{ in group } m.$$

(ii) PS_4 , Q_5

(iii) u

(iv) u

(v) u .

Math 2 Q3

Q3 (a) (i).

$$\xi(\lambda, \gamma, R) = \sum_{n=0}^{\infty} \int_{\Omega_n} \gamma^{(x, R)} \lambda^n dx_1 \dots dx_n$$

(b) (i) $\lambda \in \mathbb{R}^+$, $\gamma \in [0, 1]$, $R \in \mathbb{R}^+$

$$\pi(\lambda, \gamma, R | x) \propto \underbrace{\pi(\lambda, \gamma, R) p(x | \lambda, \gamma, R)}_Z$$

$$Z = \int_0^1 \int_0^1 \int_0^{\infty} \pi(\lambda, \gamma, R) p(x | \lambda, \gamma, R) dR d\gamma d\lambda$$

also $Pr(|x| \in \Omega_n) = Pr(x \in \Omega_n)$

$$|x| \in \Omega_n \text{ counts} \quad = \int_{\Omega_n} p(x | \lambda, \gamma, R) dx$$

$$= \xi(\lambda, 1, R) \int_{\Omega_n} \lambda^n dx$$

$$= \frac{\lambda^n \xi(\lambda, 1, R)}{n!} \underbrace{\int_{\mathbb{R}^n} dx_1 \dots dx_n}_{L=1}$$

So $Pr(x \in \Omega_n) \propto \frac{\lambda^n}{n!} \quad n = 0, 1, \dots$

$\Rightarrow |x| \sim \text{Poi}(\lambda)$.

(b) i) $y^{(t)}$ $t=1, \dots, T$ ABC samples.

Suppose $y|s \sim \mu(s) + \epsilon$

den of ϵ not depending on s (ie shift in data just mean shift).

Suppose $\mu(s') \approx \beta_1 + \beta_2(s' - s)$

for $d(s', s)$ small.

$$\Rightarrow y^{(t)} \approx \beta_1 + \beta_2(s(y^{(t)}) - s(y)) + \epsilon^{(t)}$$

$\Rightarrow$ estimate $\hat{\beta}_1, \hat{\beta}_2$ by regressing $y^{(t)}$ on $s(y^{(t)}) - s(y)$

$$\Rightarrow \tilde{\epsilon}^{(t)} = y^{(t)} - \hat{\beta}_1 - \hat{\beta}_2(s(y^{(t)}) - s(y))$$

now $\mu(s(y)) \approx \hat{\beta}_1$

$$\Rightarrow \mu(s(y)) + \epsilon^{(t)} \approx y|s(y)$$

$$\text{so } \hat{\beta}_1 + y^{(t)} - \hat{\beta}_1 - \hat{\beta}_2(s(y^{(t)}) - s(y)) \\ \approx y|s(y) \text{ approx}$$

$$\text{so } y_{adj}^{(t)} = y^{(t)} - \hat{\beta}_2(s(y^{(t)}) - s(y)) \\ \text{is better distributed } \approx y|s(y)$$

(b) i

$$\delta, \lambda, R \sim \pi(\delta, \lambda, r \mid d(X, x) < \rho)$$

where X are "unknown to us" data
and x is observed data.

It is as if we only knew the data
up to some uncertainty.

$$(c) \quad \alpha = \frac{\pi(x')}{\pi(x)} \frac{1/(n+1)}{1/|A|}$$

$$= |A| \frac{\lambda}{n+1} \delta^{C(x') - C(x)}$$

$$ii \quad \alpha = |A| \frac{\pi(x')}{\pi(x)} \frac{1/|A|}{1/n}$$

$$= |A| \frac{n \delta^{C(x') - C(x)}}{\lambda}$$

iii $X_t = x$ wp $\frac{1}{2}$ $x' \in x \cup z$ $z \in U(A)$
wp $\frac{1}{2}$ $x' = x \setminus x_i$ ($i \in U(1 \dots L)$)
wp $\alpha_{(i)}$ (if add) or $\alpha_{(i)}$ (if delete) accept
 x' (set $X_{t+1} = x'$) else reject ($X_{t+1} = x$)